

Getting API keys for the research forager

This guide is for anyone who has never requested an API key before. It takes about 15 minutes, plus waiting time for Semantic Scholar and CORE.

What is an API key?

An **API** (Application Programming Interface) is how a program asks a website for data directly, without clicking through web pages. An **API key** is a long random password that identifies *your* program to that website. It lets the site give you a fair share of access, and lets them contact you if something goes wrong.

Treat every key like a password.

- Never paste it into code, emails, chat messages or AI assistants.
- Keep it in a separate file to the code.
- If a key is ever exposed, delete it on the provider's website and create a new one.

Which keys do I need?

The forager searches nine sources. Five need no key at all; four work better, or only, with one.

Source	Key needed?	How long to get	What happens without it
OpenAlex	Yes, strongly recommended	~2 minutes, instant	Only 100 test credits a day, then requests are refused
Semantic Scholar	Optional	Days to weeks (reviewed)	Works, but shares a slow, crowded public rate limit
CORE	Yes	Usually within a day, by email	The forager skips CORE
Zenodo	Optional	~3 minutes, instant	Works, with lower rate limits
DuckDuckGo, arXiv, OSF Preprints, HAL, DOAJ	No	–	–

If you only do one, get the OpenAlex key. OpenAlex is one of the richest sources and also supplies keywords to the cloud agent. Since 13 February 2026 it has required a free key for normal use.

1. OpenAlex (free, instant)

1. Go to **openalex.org** and click **Sign up**. Create a free account with your email address.
2. Once logged in, go to **openalex.org/settings/api**.

3. Copy the key shown on that page.
4. Paste it into `API_KEYS.md` :

```
OPENALEX_API_KEY = paste-your-key-here
```

A free key gives 100,000 credits a day, far more than the forager uses.

2. Semantic Scholar (free, needs approval)

1. Go to semanticscholar.org/product/api and find the **API key request form**.
2. Fill it in with your name, email, organisation (your university is fine), and a short description of your use. For example: *"PhD literature review: an automated script that searches for recent papers on my topic, around 20–50 requests per session."*
3. Submit it and wait. Requests are reviewed by hand, so this can take anything from a few days to a few weeks. The key arrives by email.
4. Paste it into `API_KEYS.md` :

```
S2_API_KEY = paste-your-key-here
```

You don't have to wait: leave the line blank and the forager still uses Semantic Scholar, just more slowly.

3. CORE (free, emailed)

1. Go to core.ac.uk/services/api.
2. Click **Register now** (or similar) and fill in the form: name, email, organisation, and intended use. Choose the **free / academic** option.
3. The key is emailed to you, usually within a day. Check your spam folder.
4. Paste it into `API_KEYS.md` :

```
CORE_API_KEY = paste-your-key-here
```

CORE indexes full-text open-access papers from university repositories, so it's often the best source of complete PDFs.

4. Zenodo (free, instant)

1. Go to zenodo.org and click **Sign up**. You can use an email address, or log in with GitHub or ORCID.
2. Once logged in, click your **user name (top right)** → **Applications**.
3. Under **Personal access tokens**, click **New token**.
4. Give it a name such as `research-forager` . **Don't tick any scopes**, because searching public records doesn't need them.

5. Click **Create** and **copy the token immediately**. Zenodo only shows it once.
6. Paste it into `API_KEYS.md`:

```
ZENODO_API_KEY = paste-your-token-here
```

Filling in `API_KEYS.md`

When you're finished, the file should look like this (with your real keys):

```
OPENALEX_API_KEY = a1b2c3d4e5...
S2_API_KEY = s2k-XyZ...
CORE_API_KEY = Ab12Cd34...
ZENODO_API_KEY = Qw12Er34Ty...
```

Rules for the file:

- One key per line, in the form `NAME = value`.
- Keep the names exactly as they are: capitals and underscores.
- Quotation marks around the value are optional. Spaces around the `=` are fine.
- Leave a value blank to skip that service.

Checking your keys work

Run the forager and watch its output:

You see	It means
<code>[+] Fetching: ...</code> after a source's <code>Querying...</code> line	The source (and its key) is working
<code>[!] No API key file found ...</code>	<code>API_KEYS.md</code> isn't next to the notebook, or is misnamed
<code>[*] Skipping CORE: no CORE_API_KEY in API_KEYS.md</code>	The CORE line is blank
<code>... rate limit. Jittering for ...</code>	Normal; the forager is waiting politely. If it happens on <i>every</i> query, check that source's key
A source prints its <code>Querying...</code> line and then nothing	The key may be wrong or expired. Re-copy it from the provider's website

If you use Git or share the project

Add this line to a file called `.gitignore` in the project folder, so the keys are never committed:

API_KEYS.md

When sharing the project, share `API_KEYS_GUIDE.md` and a **blank** copy of `API_KEYS.md`, never your filled-in one.